Introduction

Tree

Data Structures and its Applications (UE25CS52A)

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Finding Running Median using Max-heap and Min-heap

- Step 1:
- Understand Callbacks
  - Parameter of a function makes data flexible
  - Parameter of type pointer to function makes operations flexible
  - Experiment with function programming concept
  - Write functions for
    - Map
    - Filter
    - Reduce

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Finding Running Median using Max-heap and Min-heap

- Step 2:
- Implement a flexible heap
- While initializing provide a function which decides how the comparison is made
- Implement the same heap as
  - MaxHeap
  - MinHeap

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Finding Running Median using Max-heap and Min-heap

- Step 3:
- Find running median
- Data structures:
- Max-Heap (left_half): Stores the smaller half of the numbers.
  - The root gives the largest element of this lower half.
- Min-Heap (right_half): Stores the larger half of the numbers.
  - The root gives the smallest element of this upper half.

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Finding Running Median using Max-heap and Min-heap

- Step 3 (continued):
- Find running median
- Algorithm:
- Step-by-Step Algorithm for Each Incoming Number x
- 1. Insert:
- If left_half is empty or x <= root(left_half),
- push x into left_half.
- Otherwise, push x into right_half.

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Finding Running Median using Max-heap and Min-heap

- Step 3 (continued):
- Find running median
- Algorithm:
- Step-by-Step Algorithm for Each Incoming Number x
- 2. Balance:
- If the size of left_half exceeds right_half by more than 1
- pop the root from left_half and push it into right_half.
- If the size of right_half exceeds left_half,
- pop the root from right_half and push it into left_half.

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Experiential Learning

Finding Running Median using Max-heap and Min-heap

- Step 3 (continued):
- Find running median
- Algorithm:
- Step-by-Step Algorithm for Each Incoming Number x
- 3. Query Median:
- If total elements N is even: The median is the average of the
- two root elements:
- If total elements N is odd: The median is the root of the heap
- that has the extra element.

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THANK YOU

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